TAWHID CHOWDHURY

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RESEARCH INTERESTS

Reinforcement learning for sequential decision-making; deep Q-networks and classical Q-learning for dynamic resource allocation; multi-agent scheduling; simulation of urban and combinatorial systems (SUMO); and applied machine learning for structured prediction from clinical and document data.

EDUCATION

BRAC University, Dhaka, Bangladesh January 2022 – March 2026
B.Sc. in Computer Science and Engineering *CGPA **3.73 /4.00***

- Vice Chancellor's List (1 semester); Dean's List (6 semesters); merit-based tuition scholarship, 2022–2026.
- Undergraduate thesis defended and approved by the BRAC University Thesis Committee (February 2026).

St. Joseph Higher Secondary School, Dhaka July 2018 – December 2020
Higher Secondary Certificate, Science *GPA **5.00 / 5.00***

Dhanmondi Government Boys' High School, Dhaka January 2008 – March 2018
Secondary School Certificate, Science *GPA **5.00 / 5.00***

RESEARCH EXPERIENCE

Adaptive Urban Traffic Control via Reinforcement Learning February 2026
Undergraduate thesis · Python, SUMO, Q-learning, DQN *Thesis group leader*

Supervisors: Prof. Amitabha Chakrabarty and Prof. Md. Golam Rabiul Alam, Department of CSE, BRAC University.

- Formulated dynamic signal control as a sequential decision problem and modeled volatile urban networks in the SUMO traffic simulator.
- Designed, tuned, and compared tabular Q-learning and Deep Q-Networks against fixed-time signal heuristics, with emphasis on reward design.
- In simulated congestion scenarios, reduced average vehicle waiting time by 60–70% and increased throughput by 35–40% relative to fixed-time control.
- Led the student research team for more than a year, coordinating experiments, implementation, and supervisor milestones. Manuscript based on this work is in preparation for peer-reviewed submission.

SELECTED PROJECTS

Biomarker-Based Pancreatic Cancer Diagnosis January 2026
Python, scikit-learn, XGBoost, pandas *Independent project*

- Built a non-invasive classification pipeline on 590 clinical and urinary biomarker samples, comparing SVM, Random Forest, and XGBoost.
- Handled missing values and class imbalance; evaluated models with precision, recall, F1, and ROC. Best accuracy 90.68% (XGBoost).

Bangladesh NID Extractor August 2026
FastAPI, Google Cloud Vision, OpenCV, Pillow, Docker

- Built a document-understanding pipeline that extracts structured fields from Bangladesh National ID images using OCR and neural machine translation.
- Added OpenCV/Pillow quality checks and a Dockerized inference API with JSON outputs and in-memory processing (no persistent image storage).

Nirapod: All-in-One Fast Response Service May 2025
Spring Boot, PostgreSQL, JavaScript · team of four *Team lead*

- Proposed and led a modular civic-response system covering fire service, animal welfare, and city corporation workflows.
- Implemented REST APIs for complaint CRUD and filtering. Ranked among the top two projects in the course section.

Health Guardian February 2024
HTML, CSS, PHP, MySQL · database systems project *Project lead*

- Designed ER/EER schemas and a symptom-to-diagnosis feature; wrote SQL for filtering, aggregation, and patient and ambulance reporting.

Embedded and graphics coursework 2023 – 2024
Arduino, Python, OpenGL

- Obstacle-avoiding line-following robot on Arduino with proximity sensing (about 10–15% fewer collisions in physical tests). 2D game rendered from OpenGL points to limit primitive overhead.

TECHNICAL SKILLS

Programming. Python, C, C++, Assembly, Verilog; JavaScript (basic).
Machine learning. Q-learning, DQN, SVM, Random Forest, XGBoost; scikit-learn; supervised classification and evaluation (precision, recall, F1, ROC).
Simulation and vision. SUMO; OpenCV, Pillow; Google Cloud Vision OCR; document preprocessing.
Data and systems. pandas, Matplotlib, Excel, Google Sheets; FastAPI, Spring Boot, REST; MySQL, PostgreSQL; Docker, Git, LaTeX, Google Colab.
Hardware. Raspberry Pi 5, Arduino. Basic exposure to Postman, JMeter, and Playwright.

HONORS AND AWARDS

- Vice Chancellor's List, BRAC University — highest academic distinction for exceptional GPA (1 semester).
- Dean's List, BRAC University — 6 semesters.
- Merit-based scholarship, BRAC University, 2022–2026.
- Physics Brawl, Open Category — ranked 104th globally; team lead, Physics Olympiad.
- Mathematics Olympiad recognition and perfect attendance, St. Joseph Higher Secondary School.

LEADERSHIP AND SERVICE

- **Organization and Management Executive, Bizbee Business Club, BRAC University (2022–2026).** Ran recruitment for the department and coordinated logistics for campus business events.
- **Joint Secretary, author, and editorial member, Josephites Math Club (2018–2020).** Managed logistics for a digital presentation competition (500+ participants) and the editorial pipeline for the annual Summation magazine.
- **Student Cabinet leader, Dhanmondi Government Boys' High School (2015–2018).** Elected to represent 800+ students. Founded the school Rubik's Cube Club and designed training sessions for 20+ members.

CERTIFICATIONS

- Fully Automated MLOps;
- Developing Machine Learning Models for Production;
- MLOps Deployment and Life Cycle — DataCamp, September 2025.

ENGLISH LANGUAGE PROFICIENCY

IELTS Academic 3 October 2026

Test Report Form 26BD511212CHOT050A · Centre BD050 · Candidate 511212 CEFR C1

- Overall band 8.0. Listening 8.5; Reading 8.0; Writing 7.5; Speaking 7.5.
- No band less than 7.5

Other languages: Bangla (native); Hindi (conversational).

REFERENCES

Dr. Amitabha Chakrabarty, Professor, Department of CSE, BRAC University (Thesis supervisor). Email

Dr. Md. Golam Rabiul Alam, Professor, Department of CSE, BRAC University (Thesis supervisor). Email

Dr. Swakkhar Shatabda, Professor, Department of CSE, BRAC University. Email